

Guillermo Serrera
guillermo.serrera@unican.es
Av. Los Castros 48 Santander 39005 Spain

Employment

2023– **Predoctoral researcher**, University of Cantabria

Education

University education

2022–26* **Ph.D.**, Science and Technology, University of Cantabria
2021–22 **M.Sc.**, Science and Engineering of Light, University of Cantabria
2016–21 **B.Sc.** Physics, University of Cantabria

Other training

2025 ACS Reviewer Lab, American Chemical Society
2025 Optica Publishing Group Certified Reviewer Course, Optica Society of America
2025 Fundamentals and applications of optical materials, University of Cantabria summer school
2024 Certified Peer Reviewer Course, Elsevier Researcher Academy
2023 Creative strategies in teaching and research, University of Cantabria courses
2023 Social competences: tools for university teachers, University of Cantabria courses
2023 Emotional competences: tools for university teachers, University of Cantabria courses
2022 European School on Plasmonics and Phase Change Materials, PHEMTRONICS.

Research Experience

2023– Researcher in the project "Modelling of the Photothermal response of hybrid nanoSystems (MOPHOSYS)" Advisor: Pablo Albella (University of Cantabria), Spain.
2021–2022 Part time researcher in the project "Spectroscopy of planetary and atmospheric particulae by optical tweezers". Advisor: Onofrio Maragò (IPCF-CNR), Italy.

Awards, Funding & Honors

2025 Funding for short research stays (EST25/00316).
2025 Best theoretical poster at CEN2025 conference.
2023 Best poster at University of Cantabria Science day.

*Expected.

- 2023– University Professorship Training (FPU21/02296) Fellow at the University of Cantabria.
- 2020-2021 Collaboration Research Grant at the Applied Physics Department of the University of Cantabria.

Publications

 [Google Scholar](#)

Journal Articles

- J1. **Serrera, G.**, Tanaka, Y. Y. & Albella, P. 3-dimensional plasmonic nanomotors enabled by independent integration of optical pulling and lateral forces. *Nanophotonics* **14**. <https://doi.org/10.1515/nanoph-2025-0374> (2025).
- J2. Siegel, N., Sanz-Paz, M., González-Colsa, J., **Serrera, Guillermo**, Zhu, F., Szalai, A. M., Kołataj, K., Fujii, M., Sugimoto, H., Albella, P. & Acuna, G. P. Distance-Dependent Interaction between a Single Emitter and a Single Dielectric Nanoparticle Using DNA Origami. *Small Structures*. <https://doi.org/10.1002/sstr.202500299> (2025).
- J3. **Serrera, G.**, González-Colsa, J. & Albella, P. Amplified linear and nonlinear chiral sensing assisted by anapole modes in hybrid metasurfaces. *Applied Physics Letters* **124**. <https://doi.org/10.1063/5.0212393> (2024).
- J4. González-Colsa, J., **Serrera, G.**, Saiz, J. M., Ortiz, D., González, F., Bresme, F., Moreno, F. & Albella, P. Gold nanodoughnut as an outstanding nanoheater for photothermal applications. *Optics Express* **30**. <https://doi.org/10.1364/OE.446637> (2022).
- J5. **Serrera, G.**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. Enhanced optical chirality with directional emission of Surface Plasmon Polaritons for chiral sensing applications. *Journal of Quantitative Spectroscopy and Radiative Transfer* **284**. <https://doi.org/10.1016/j.jqsrt.2022.108166> (2022).
- J6. González-Colsa, J., **Serrera, G.**, Saiz, J. M., González, F., Moreno, F. & Albella, P. On the performance of a tunable grating-based high sensitivity unidirectional plasmonic sensor. *Optics Express* **29**. <https://doi.org/10.1364/OE.422026> (2021).

Presentations

Talks

- T1. **Serrera, G.**, Tanaka, Y. Y. & Albella, P. *Control of 3-Dimensional Motion in Plasmonic Nanomotors with Optical Pulling Forces* Invited talk at META 2025 (Málaga, Spain). July 2025.
- T2. **Serrera, G.**, Tanaka, Y. Y. & Albella, P. *Optical pulling forces for 3-dimensional motion control in plasmonic nanomotors* 13th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2025) (Porto, Portugal). Feb. 2025.
- T3. **Serrera, G.**, González-Colsa, J. & Albella, P. *Anapole modes enabling linear and nonlinear chiral sensing in hybrid metasurfaces* META 2024 (Toyama, Japan). July 2024.
- T4. **Serrera, G.**, González-Colsa, J. & Albella, P. *Exploiting Hybrid Metasurfaces for enhanced Linear and Nonlinear Chiral Sensing Applications* 12th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2024) (Rome, Italy). Feb. 2024.

- T5. **Serrera, G.**, González-Colsa, J. & Albella, P. *Hybrid metal-dielectric resonator for anapole-assisted enhanced chiral sensing applications* SPIE Photonics Europe (Strasbourg, France). Apr. 2024.
- T6. **Serrera, G.**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. *Chiral sensing based on directional plasmons* International Conference of Quantum, Nonlinear and Nanophotonics 2022 (ICQNN'22) (Jena, Germany). Sept. 2022.
- T7. **Serrera, G.**, González-Colsa, J., Giannini, V., Saiz, J. M. & Albella, P. *Directional Plasmon Emission in a Mixed Dielectric-plasmonic Structure: Application to Chiral Sensing* 10th International Conference on Photonics, Optics and Laser Technology (PHOTOPTICS 2022) (Online). Sept. 2022.

Posters

- P1. **Serrera, G.**, Tanaka, Y. Y. & Albella, P. *Innovative approaches to 3D motion control in plasmonic nanomotors with optical pulling forces* CEN2025 (Madrid, Spain). June 2025.
- P2. **Serrera, G.**, González-Colsa, J., González, F., Moreno, F., Saiz, J. M. & Albella, P. *Enhanced chiroptical activity with dielectric chiral nanostructures: applications in racemic-mixture sensing* International Conference on Metamaterials and Nanophotonics (Metanano 2021) (Online). Sept. 2021.
- P3. **Serrera, G.**, González-Colsa, J., González, F., Moreno, F., Saiz, J. M. & Albella, P. *Grating metasurfaces as directional plasmon sources: applications in achiral and chiral sensing* Imagine-Nano 2021 - 3PM2021 (Bilbao, Spain). Nov. 2021.

Teaching

University of Cantabria

2025–	Teaching Assistant, Physics I (G375)
2025–	Teaching Assistant, Physics II (G383/G393)
2024–	Teaching Assistant, Advanced Experimental Techniques (G1775)
2023	Teaching Assistant, Energy in the World of Today (G1675)

Supervisions

Undergraduate

2024	Bárbara Cacicedo López, University of Cantabria, <i>Study of structured light scattering by nanostructures: application in optical chirality</i>
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Research stays and visits

2025	University of Fribourg, two-month research stay at the Photonic Nanosystems Group, supervised by Prof. Guillermo P. Acuña.
2024	Ghent University, three-month research stay at the Photonics Research Group, supervised by Prof. Alberto G. Curto.

- 2021 Instituto de Estructura de la Materia, CSIC, visit to Dr. Vincenzo Giannini, assisted to the seminars "Novel plasmonic metamaterials: applications in sensing", and "Thermal effects in nanoparticles".
- 2020 University of Oviedo, visit to Prof. Pablo Alonso-González, assisted to the seminars "Advanced near-field optics".

Last updated: November 2, 2025